

Human in the loop: what AI may and may not decide

The decision boundary, the tests that generate it, and the authority mechanics that make it hold

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— Summary

"Human in the loop" is an authority statement, not a posture. This document distinguishes the three positions a human can actually hold, gives five tests that place any controls decision on the human side of the line, sets out the decision boundary those tests generate, and covers the mechanics that make it hold in practice — delegated authority thresholds, what a signature means, how rubber-stamping starts and is detected, and what changes when AI runs several steps in a chain.

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Human in the loop: what AI may and may not decide

In one paragraph. "Human in the loop" is an authority statement, not a posture. This document distinguishes the three positions a human can actually hold, gives five tests that place any controls decision on the human side of the line, sets out the decision boundary those tests generate, and covers the mechanics that make it hold in practice — delegated authority thresholds, what a signature means, how rubber-stamping starts and is detected, and what changes when AI runs several steps in a chain.

Who this is for. Project controls managers, cost and planning leads, change board members and project directors who hold delegated authority, and the practitioners whose names go on AI-assisted outputs.

— 1. The phrase does no work on its own

"There is a human in the loop" is offered as an assurance in almost every AI deployment discussion, and on its own it assures nothing. A human reading a screen is in the loop. A human clicking "approve" on forty items in six minutes is in the loop. A human who could not reproduce the number if asked is in the loop.

The phrase becomes a control when it answers three questions: **which decisions** does the human make rather than confirm; **at what threshold** does the decision move to someone more senior; and **what must the human have done** before their signature is worth anything. This document answers those three, in that order.

The discipline-specific boundaries are owned elsewhere in this series: [AIG-04 – AI-assisted cost forecasting](#), [AIG-05 – AI in scheduling](#) and [AIG-06 – AI for risk identification and quantification](#). What follows is the general apparatus that generates those lists, and the authority mechanics none of them covers.

— 2. Three positions, only one of which is what people mean

Position	What it means operationally	Where it is acceptable
In the loop	The system proposes; a human evaluates and decides; nothing proceeds without that decision	Any output meeting a test in §3
On the loop	The system acts; a human monitors, samples and can intervene	High-volume, low-consequence work with a measured error rate and an active monitoring duty (Tier 3, AIG-08 §4)
Out of the loop	The system acts; no human reviews	Only where the action is trivially reversible with no external reliance or contractual effect — in controls, almost nothing

The distinction is not philosophical. In the loop costs review time per item and is the only position that gives you a decision-maker. On the loop costs monitoring effort and gives you a detection lag: errors happen, and you find them at the sampling frequency. Choosing "on the loop" accepts a known error rate — a legitimate professional choice, and only legitimate when the rate has been measured and the acceptance recorded.

Most deployments described as "human in the loop" are, on inspection, on the loop. That is often the right design. It becomes a governance failure when the framework claims the first and operates the second, so the monitoring duty that makes "on the loop" safe is never resourced.

— 3. Five tests that place a decision

Apply these to any controls decision an AI tool touches. **One positive answer puts the decision on the human side of the line.** They are deliberately blunt, because a boundary that requires fine judgement to apply will not survive a busy month.

1. Is it irreversible within the reporting cycle? A baseline changed, a contingency released, a notice served, a payment certified — none can be quietly undone next period. Reversibility, not size, is the first discriminator.

2. Does it have contractual or legal effect? Anything that asserts, concedes, waives or triggers a right. See [AIG-07 – AI in document control and correspondence](#) §3 for how this plays out in writing.

3. Will someone outside the project rely on it? A board, client, lender, auditor or certifier. External reliance converts an internal analysis into a representation, and representations are made by people.

4. Is the question fundamentally why, and contested? Attribution of cause — who caused the delay, whether escalation was foreseeable, whether rework is a defect or a variation — is the substance of most disputes. A model can assemble the evidence; it cannot hold a position.

5. Does it affect a person? Allocation, performance assessment, selection, capability judgement. This attracts data-protection and employment obligations that vary by jurisdiction, and a decision made about someone by a system nobody can explain is indefensible in every one of them.

One gating condition sits on top of the tests: **an output that cannot be reconstructed cannot carry any of these decisions**, whatever its apparent quality — see [AIG-09](#) §8.

— 4. The boundary

The tests generate the following. This is the function-level list; the discipline documents refine it.

Decision	AI may	A competent human must
Baseline change	Assemble the change pack, quantify time and cost impact, check the register for related items	Decide whether it is approved, on what scope and value, and sign the baseline revision (BPG-04)
Contingency release	Match the drawdown to the risk that materialised, compute remaining balance and burn rate	Decide the release, its amount and the authority level it requires (BPG-10 , §5)
Extension of time entitlement	Assemble the chronology, extract notices and windows, run delay analyses on stated assumptions	Decide the entitlement position, the analysis method, and what is claimed or resisted (BPG-12)
A forecast submitted to a board or client	Produce candidate forecasts, decompose movement, flag inconsistency with the schedule	Choose the basis, state the assumption, own the number and sign (AIG-04)
Revenue recognition, provisioning, accrual judgement	Draft the working, run consistency checks, map facts to the policy	Make the judgement against the standard and the entity's policy, with the accounting owner
Acceptance of a subcontractor's programme or valuation	Check logic, compare against contract dates, reconcile quantities	Accept, reject or accept with reservation — acceptance is a commercial act
The contingency level adopted	Run the simulation, rank drivers, present the distribution	Choose the confidence level consistent with the organisation's appetite, and justify it (AIG-06)
What is disclosed, and how it is framed	Draft, check internal consistency, flag omissions	Decide what is said, what is caveated and what is not said
Decisions about people	Nothing that scores or ranks individuals without a specific, lawful, explainable basis	Decide, on evidence they can explain to the person affected
Changing a rule of credit or measurement basis	Model the effect of the change	Decide it, and disclose the discontinuity in the reported trend

Two things this table is not. It is not a list of tasks to keep AI away from — every row has a substantial AI contribution in the middle column, and refusing that contribution is its own failure. And it is not a maturity statement that relaxes as tools improve: the tests in §3 concern the nature of the decision, and a more capable model does not make a contested entitlement position less contested.

— 5. Thresholds: writing authority so it binds

A boundary without thresholds collapses into either paralysis or nothing. Delegated authority is the mechanism, and three rules make it work with AI in the picture.

Use the project's existing schedule. Do not invent an AI-specific authority ladder. AI-assisted decisions run through the same delegated authority as any other, expressed the same way — a value, a percentage of budget at completion (BAC), or a category. A parallel ladder guarantees divergence and gives an approver two answers to choose between.

Express the threshold on the decision, not the tool. "Contingency releases up to 0.25 % of BAC: controls manager" is a control. "AI-assisted releases: controls manager" is not, because it invites splitting a decision into an AI-assisted part and a human part to land in a lower band.

Name the aggregation rule. Thresholds are defeated by division. Related releases within a period, or against the same risk, aggregate for the threshold — and the rule is written down, because the split is rarely deliberate and always available. §10's worked example turns on this.

Add one AI-specific provision, and only one: **the approver must be able to state the basis of the proposal in their own words.** If they cannot, it is not ready for approval — a rule that is cheap to apply and catches the rubber-stamp before it becomes habitual.

— 6. What a signature actually asserts

A signature on an AI-assisted output asserts four things, and these are what it will be tested against.

1. **The inputs came from source**, not from the model's restatement of them.
2. **The method is known and appropriate**, and the reason for choosing it can be stated.
3. **The assumptions are stated** beside the answer, so a reader can challenge them.
4. **Differences between the model's output and the signed output are recorded**, with reasons.

"The model produced it" is not a defence, and neither is "the tool is approved" — the register governs which tools may be used and says nothing about whether this output is right. The verification record of [AIG-09](#) §3 is what the signature rests on.

The corollary is uncomfortable and worth stating plainly: **a reviewer who could not have produced the output by another route is not in a position to sign it.** Not every signatory must redo every calculation, but the capability must exist in the review, and where it does not the output goes to someone who has it. This is why the deskilling problem in [AIG-12 – The AI-literate controls professional](#) is a governance problem, not a training-department problem.

— 7. How rubber-stamping starts

It does not start with negligence. It starts with a run of correct outputs. The tool is accurate for several cycles, review finds nothing, and review time falls because nothing rewards it. The reviewer begins checking that the output looks reasonable rather than that it is right — a much faster operation, indistinguishable from the outside. Then an output is wrong in a way that looks reasonable, and it passes.

Three detections, none expensive.

Review time per output. If time booked to verification has fallen materially while volume has not, that is the signal. It is measurable if verification is a costed line ([AIG-08](#) §10), invisible if not.

Change rate. If no AI-assisted output has been materially changed in review for several cycles, either the tool is excellent or the review is not happening. Both are worth knowing; only one is being assumed.

Spot reconstruction. Once a quarter, pick a signed output at random and ask its signatory to reconstruct it (AIG-09 §1). It is the only test that distinguishes the two cases above, and it takes under an hour.

— 8. When AI runs several steps

Chained and agentic systems — where the tool retrieves the data, computes, drafts and assembles without stopping — change the shape of the control, not the principle.

A single-step tool that errs produces one wrong output and one verification catches it. A chain that errs at step two carries the error forward, and every later step builds on it plausibly. The error is not merely propagated; it is polished. By the final output, the workings that would have exposed it are gone.

The response is to move verification **from per-output to per-workflow**. The professional assures the design of the chain — which steps are permitted, which data each may touch, where it must stop — and inserts checkpoints where a consequential intermediate result is inspected before the chain continues. Sensible checkpoints in controls: after extraction and before computation (is this the right cut?), after computation and before narrative (do the numbers reconcile?), and before anything leaves the function.

The audit trail must now record the chain, not just the answer. The decisions in §4 do not move: a chain may prepare a contingency release pack end to end, and the release is still decided by the person with the authority.

— 9. How this goes wrong

The loop with no decision in it. The workflow includes a human step, and that step is confirmation rather than decision. Nobody has ever declined. The control exists on the diagram.

Authority set on the tool rather than the decision. The threshold reads "AI-assisted analyses require manager approval". A decision is split into a model-produced analysis and a human-entered figure and lands below the line without anyone intending it.

Thresholds defeated by division. Four contingency releases of 90,000 against the same risk in one month, each within a lower authority. No aggregation rule, no visibility, no decision at the level the total warranted.

Signature without capability. The signatory cannot recompute and has no access to someone who can within the reporting timetable. The signature is a formality and everyone involved knows it.

Boundary rewritten by capability claims. A tool improves, and the argument follows that entitlement or recognition judgements can now be automated. The tests in §3 concern the nature of the decision: nothing about a better tool makes a contested cause uncontested.

Reasonableness substituted for correctness. Six good cycles, then review becomes a glance — invisible until an output is wrong and reasonable at the same time.

Chains verified only at the end. The final pack is checked carefully and every intermediate step trusted. An error introduced at extraction is confirmed by everything built on it.

Escalation with no route. The framework says contested items escalate; nothing says to whom, by when, or what happens if the reporting deadline arrives first. Under pressure the item is signed.

— 10. Worked example — a contingency release and the authority it needs

Illustrative figures. One project, one month. Currency USD. Thresholds expressed as a percentage of budget at completion.

Setup. Project BAC is **USD 48,000,000**. Contingency held is **USD 2,400,000** (5.0 % of BAC: $2,400,000 \div 48,000,000 = 0.05$). The delegated authority schedule for contingency release reads: controls manager up to **0.25 % of BAC**; project director up to **1.00 % of BAC**; sponsor above that. An AI-assisted analysis matches a materialised risk to a proposed release of **USD 180,000** and produces the supporting pack.

Step 1 — convert the thresholds to money.

- Controls manager = $0.25 \% \times 48,000,000 = 0.0025 \times 48,000,000 = \text{USD } 120,000$
- Project director = $1.00 \% \times 48,000,000 = 0.01 \times 48,000,000 = \text{USD } 480,000$

Step 2 — place the decision. $180,000 > 120,000$ and $180,000 \leq 480,000$, so the release is the **project director's decision**, not the controls manager's.

Step 3 — apply the aggregation rule. Two releases of **USD 95,000** and **USD 70,000** were made earlier in the same month against the same risk. Aggregated: $95,000 + 70,000 + 180,000 = \text{USD } 345,000$. This is still within the director's band ($345,000 \leq 480,000$) but it is now $345,000 \div 2,400,000 = 14.4 \%$ of the total contingency drawn against a single risk in one month — a fact the director should be told and would not have been told by three separate approvals.

Step 4 — what AI contributed. The tool matched the drawdown to the risk, computed the remaining balance ($2,400,000 - 345,000 = \text{USD } 2,055,000$) and the burn rate, and assembled the evidence. It did not decide that the risk had materialised, that the amount was right, or that the remaining contingency is adequate for the risks that have not.

Result. A director-level decision, made on an aggregated view, with a remaining contingency of **USD 2,055,000** against a residual exposure the director must now judge.

Interpretation. The arithmetic is trivial; the control is not. Without the aggregation rule of \$5, three defensible individual approvals produce an undisclosed concentration — each approver with authority, nobody seeing the total. Note also what the example refuses to do: treat the model's confidence in the match as a reason to lower the authority level. The threshold is a property of the decision's consequence, and the consequence of an over-drawn contingency does not change with how the recommendation was produced.

— 11. Checklist

- [] The framework states, per use, whether the human is in the loop or on the loop — and where it is on the loop, the error rate is measured and monitoring has a named owner.
- [] Every decision an AI tool touches has been run through the five tests in §3.
- [] The boundary in §4 has been adapted to this project's decisions and written into the plan.

- [] Thresholds are expressed on the decision rather than on AI involvement, and use the project's existing delegated authority schedule.
- [] An aggregation rule exists for related decisions within a period.
- [] Approvers can state the basis of a proposal in their own words.
- [] Signatories know the four assertions in §6, and outputs go to someone who can verify them when the first reviewer cannot.
- [] Review time and change rate are visible, and a spot reconstruction is done each quarter.
- [] Chained workflows have named checkpoints, and the audit trail records the chain.
- [] Escalation names a person and a deadline, and says what happens when the reporting date arrives first.

The line is not drawn to keep AI out of the work. It is drawn so that when a decision is questioned, there is a person who made it, on grounds they can state — which is the only thing that has ever made a project controls number worth anything.

— Related

- [AIG-08 – Governing AI on a project – the control framework](#) — the register, tiers and records this boundary is enforced through
- [AIG-09 – Bias, explainability and auditability](#) — why an unreconstructable output cannot carry a decision at all
- [AIG-04 – AI-assisted cost forecasting](#) — the forecasting-specific boundary these tests generate
- [BPG-04 – Baselining and baseline change control](#) — the change authority this document assumes exists
- [BPG-10 – Contingency and management reserve](#) — the underlying discipline behind the worked example

— Sources and standards

- PCI Body of Knowledge, Domain 12 (Risk Management for Project Controls) and Domain 13 (AI for Project Controls & Project Management), [docs/bok/](#) — explained in our own words, not reproduced.

Delegated authority schedules, accounting judgements and obligations relating to decisions about people are governed by an organisation's own policies and by law that varies between jurisdictions. This document describes the controls discipline and is not legal or accounting advice.

— Status and version

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